Example 5.11 The weights of 10 students (in *kg*) are 35, 42, 40, 38, 25, 32, 29, 45, 20, 24 Find the median of their weight?

Solution

Arranging the weights in ascending order, we have,

20, 24, 25, 29, 32, 35, 38, 40, 42, 45

Here, $n = 10$, which is even.

$$\text{Therefore, median weight} = \frac{1}{2}\left\{\left(\frac{n}{2}\right)^{th} term + \left(\frac{n}{2}+1\right)^{th} term\right\}$$

$$= \frac{1}{2}\left\{\left(\frac{10}{2}\right)^{th} term + \left(\frac{10}{2}+1\right)^{th} term\right\}$$

$$= \frac{1}{2}\left\{5^{\text{th}} \text{ term} + 6^{\text{th}} \text{ term}\right\}$$

$$= \frac{1}{2}\{32+35\}\ kg = \frac{67}{2} = 33.5\ kg$$

Hence, Median is 33.5 *kg*.

Example 5.12 Create a collection of 12 observations with median 16.

Solution

As the number of observations is even, there are two middle values.

The average of those values must be 16.

We will now find any pair of numbers whose average is 16. Say 14 and 18.

Now an example of data with median 16 can be 2, 4, 7, 9, 12, 14, 18, 24, 28, 30, 45, 62.

Note

We can find more than one answer for this question

Example 5.13 The lifetime (in days) of 11 types of LED bulbs is given in days. 365, 547, 730, 1095, 547, 912, 365, 1460, 1825, 1500, 2000. Find the median life time of the LED bulbs.

Solution

Arranging the data in ascending order, we have,

365, 365, 547, 547, 730, 912, 1095, 1460, 1500, 1825, 2000.

The number of observations are 11, which is odd.

$$\text{Median} = \left(\frac{n+1}{2}\right)^{th} \text{term}$$

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